Sequence Modelling Basics

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1. Evaluation metrics

The model was evaluated on a binary sentiment classification task, which was made using a GRU sequential model with 128 hidden units and then two additional layers, one with 64 nodes and ReLU activation, and the last output layer with sigmoid activation.

1.1 Accuracy

$$\text{Accuracy} = \frac{\text{TP} + \text{TN}}{\text{Total}} = \frac{432 + 338}{1000} = 0.77$$

1.2 Precision, Recall, and F1-Score

Class	**Precision**	**Recall**	**F1-Score**	**Support**
0 (Negative)	0.73	0.86	0.79	504
1 (Positive)	0.82	0.68	0.75	496
Macro Average	0.78	0.77	0.77	1000
Weighted Average	0.78	0.77	0.77	1000

Table 1: Classification Report

2. Confusion Matrix

For reference:

- True Negatives (TN): 432
- False Positives (FP): 72
- False Negatives (FN): 158
- True Positives (TP): 338

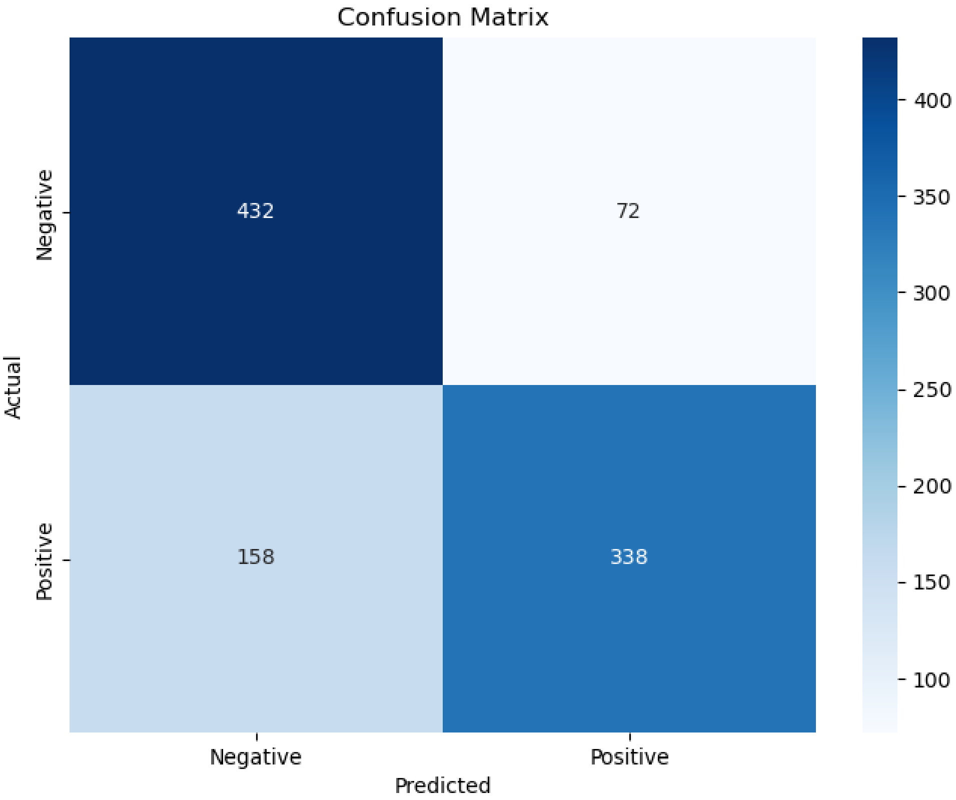

Figure 1: Confusion Matrix Visualization

3. Error Analysis

3.1 False Positives (FP = 72)

Instances where the model predicted a positive sentiment, but the ground truth was negative.

Insights:

- Errors arise due to *ambiguous tone* or *sarcasm* that appears linguistically positive but is semantically negative.
- Example: *"Great, just what I needed—another app that crashes daily."*

3.2 False Negatives (FN = 158)

Instances where the model predicted negative sentiment, but the ground truth was positive.

Insights:

- Subtle expressions of satisfaction or neutral-sounding praise were misclassified.
- Example: *"It did the job decently."*
- Suggests a need for more nuanced semantic representation in the model.

4. Minimal Clue Challenge

Review	Predicted	Actual	Reason	How to Remove
This soundtrack is my favorite music of all time, hands down. The intense sadness of "Prisoners of F...	2	1	The model is fed only the first 100 words and in this the model could sense more of the positive words than the negative ones.	Adding more words.
I was excited to find a book ostensibly about Muslim feminism, but this volume did not live up to th...	1	2	The model saw the words "did not live up to" and predicted it as negative.	Same as above.
This book is worth to keep in your collection as it does not only advise what to do with sourdough b...	1	2	The model saw the phrase "does not" and predicted it as negative.	Same as above.
This is the WORST CD-ROM game I have ever seen! My 7-year-old daughter longed for this game all Chri...	2	1	The model saw the word "longed" as positive and missed strong negative cues like "worst".	Training it for a longer time.

Table 2: Examples of misclassifications due to minimal contextual clues or ambiguity.

5. Conclusion

While the model shows a moderate performance with 77% accuracy and reasonably balanced F1-scores, the following conclusions can be drawn:

- Performance suffers in the presence of **sarcasm and indirect language**.
- There is a noticeable skew toward false negatives, implying an overcautious classification boundary.
- Enhancing the model using **transformer-based embeddings** and incorporating a **sarcasm detection module** could mitigate these errors.

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